Joint Sensing-Communication Performance Analysis of ISAC-Enabled VCN

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***Abstract*—Integrated sensing and communication (ISAC) is emerging as a key technology and research focus for future vehicular communication networks (VCN). It achieves efficient reuse of wireless infrastructure and spectrum resources through the collaborative design of sensing and communication functionalities. However, this integration leads to inevitable mutual interference, and the complexity of channel conditions further complicates the coordination of these functionalities. This paper primarily focuses on the joint sensing-communication performance analysis of ISAC-enabled VCN. Specifically, we model the spatial distribution of roads using a Poisson line process, while the locations of vehicles and roadside units (RSUs) are represented by two one-dimensional Poisson point processes. We characterize the dynamic interference distribution caused by RSUs during the sensing and communication phases and calculate the probability of successful perception (PSP) for a typical pair. Furthermore, for this typical pair, we meticulously derive the communication coverage probability based on the derived PSP for such a pair. To provide a detailed analysis of the interaction between communication and sensing functionalities, we evaluate their trade-off relationship and derive the joint probability of ISAC coverage. Moreover, we perform comprehensive simulations to verify the theoretical results. Additionally, the numerical results demonstrate how different parameters impact the network performance, providing guidance for network deployment and resource allocation under certain performance requirements.**

***Index Terms*—Integrated sensing and communication, vehicular communication networks, sensing-communication trade-off, Poisson line process.**

I. Introduction

WITH the accelerated development of intelligent transportation systems (ITS), vehicular communication networks (VCN) have become a critical means to enhance the road safety and traffic efficiency. However, as the scale and complexity of vehicular networks continue to grow, the analysis of VCN performance has become increasingly necessary and important [1], [2], [3]. As a key technology in such networks, integrated sensing and communication (ISAC) significantly improves the spectrum efficiency by integrating radar sensing and communication functionalities into a single system. With the increasing complexity of communication scenarios in VCN, ISAC plays a crucial role in enhancing communication reliability. Especially in situations where spectrum resources are limited, ISAC further improves the system performance. ISAC technology enables cooperative communication between vehicles and infrastructure by enhancing data sharing capabilities and improving synchronization between sensing and communication functions. This provides crucial technical support for advancements in autonomous driving, road safety, and traffic efficiency, etc. Consequently, ISAC has been regarded as a core technology in future VCN, driving the continuous development of ITS [4], [5].

Despite these advantages, the dense deployment of ISAC nodes in VCN presents several challenges and issues that remain to be addressed, such as conflicts and utilization efficiency of spectrum resources, the trade-off between sensing and communication performance, hardware integration design, as well as security and privacy concerns [6], [7]. In this study, we focus on analyzing the communication coverage under the condition of successful sensing and explore a joint success probability model for sensing and communication, aiming to evaluate the practical performance of ISAC-enabled VCN.

Network integration leads to an increasing overall complexity, while dynamic variations in network space and channel fading further exacerbate the difficulty of statistical performance analysis. As a modeling methodology in stochastic geometry, Poisson point process (PPP) has been widely used for modeling and analyzing the performance of large-scale networks. In wireless communication networks, specifically, one-dimensional and two-dimensional PPPs provide theoretical support for the assumption of randomly distributed nodes [8], [9]. However, the limitation of PPP lies in its inability to capture the spatial dependencies between nodes, making it difficult to accurately calculate the interference, especially in scenarios where nodes are clustered [10]. Although Poisson cluster process (PCP) performs well in modeling clustered nodes, particularly in scenarios with densely distributed base stations [11], [12], it has certain limitations when dealing

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with spatial constraints in practical networks. When applied to road traffic networks, PCP fails to accurately capture the relationship between node positions and road structures [13]. To address the aforementioned limitations, the Poisson line process (PLP) has been proposed to better describe the node distribution characteristics in vehicular networks. The nodes in vehicular networks, such as vehicles and roadside units (RSUs), are typically distributed uniformly along roads, forming a specific spatial structure. Unlike general random networks, the node distribution in vehicular networks is constrained by the geometry of the road system. Therefore, the PLP, a stochastic process tailored for line geometries, is more effective in capturing the spatial distribution of vehicles and infrastructure in vehicular networks [1], [14], [15].

Given the applicability of PLP in modeling VCN, we adopt PLP to model the road distribution in VCN in this work. Specifically, we evaluate the joint probability of ISAC coverage (JPISAC) in ISAC-enabled VCN. To achieve the coverage performance, we introduce a resource allocation strategy, in which the density ratio of sensing RSUs and communication RSUs is determined through time-division multiplexing. To validate our proposed model, we conducted extensive simulations and numerical experiments. The results show a high level of consistency between the simulations and numerical experiments.

II. Related Works

ISAC technology in VCN has offered remarkable dual capabilities, enabling vehicles and infrastructure to concurrently perform communication and sensing tasks. This integration not only reduces the reliance on standalone systems, but also enhances vehicle-to-infrastructure (V2I) communication capabilities by leveraging the echoes of communication signals for sensing purposes [16], [17]. Specifically, vehicles equipped with ISAC utilize advanced sensing functionalities to accurately detect and identify surrounding objects, encompassing other vehicles, pedestrians, obstacles, etc. Critically, they transmit vital information to nearby vehicles and RSUs, expediting decision-making processes and mitigating collision risks [5], [18]. Furthermore, ISAC technology excels in precisely tracking vehicle positions and movements. Its continuous state estimation diminishes communication overhead and latency, making it ideally suited for highly time-sensitive applications such as autonomous driving and cooperative vehicle platooning. This feature ensures seamless and efficient operations, even in dynamic and complex traffic environments [16], [19], [20].

While ISAC technology has introduced unprecedented interactive capabilities to VCN, stochastic geometry provides robust mathematical support for understanding and analyzing the spatial characteristics and the dynamic behavior of these complex networks. By employing tools such as the PPP, stochastic geometry effectively modeled the spatial distribution of vehicles and transmitters on roads, particularly in high-density and dynamic environments, capturing the complex dynamics of vehicular communications [21]. Stochastic geometry was also used to evaluate the communication performance in line-of-sight (LOS) and non-line-of-sight (NLOS) scenarios, making it particularly suitable for applications like millimeter-wave communications and autonomous driving, which demand high LOS accuracy [22]. In network performance analysis, stochastic geometry aids in studying resource selection processes and key performance indicators in vehicular networks, such as resource exclusion rates, successful transmission probabilities, and packet reception times. Through the utilization of stochastic geometry-based mathematical models, researchers derived closed-form expressions, allowing the performance of resource selection schemes to be evaluated under a wide range of parameters [23]. Additionally, stochastic geometry was employed to evaluate the interference between V2I and vehicle-to-vehicle (V2V) communications, optimizing resource allocation, reducing communication delays and interference, and further enhancing overall network performance [24].

Within the realm of stochastic geometry, the homogeneous PPP and PCP serve as prevalent tools for dissecting the distribution of vehicular nodes and their subsequent influence on network performance metrics. These methods typically assume that nodes are randomly and uniformly distributed in space, offering a theoretical basis for evaluating metrics such as coverage probability and achievable transmission rates [13], [25]. Nevertheless, while homogeneous PPP excels at modeling the uniform random distribution prevalent in generic networks, its application to actual road networks poses challenges. In these complex environments, the locations of vehicles are intricately influenced by road structures, traffic flows, and geographical factors, all of which make it difficult for the homogeneous PPP method to accurately mirror the actual distribution of vehicle nodes [13], [26]. For networks with clustering characteristics, such as the distribution of vehicles in traffic flows, advanced methods like the PCP were introduced to capture these clustering behaviors. However, PCP also struggles to accurately describe the complex spatial relationship between vehicles and road networks [25], [27].

To address these limitations, the PLP has emerged as an advanced stochastic geometry method. PLP modeled road systems as random line processes. Researchers used two one-dimensional PPPs along these lines to simulate the distribution of vehicles and RSUs, thereby more accurately capturing the spatial coupling between vehicles and road networks [1]. In practical applications, PLP provides new tools for several key areas. For example, the PLP model predicted macrocell base station load, optimized resource allocation, and improved system utilization [28]. When applied to secrecy analysis, the PLP model, combined with artificial noise techniques, enhances physical layer security, making it particularly suitable for high-security vehicle-to-everything communications [29]. When it came to the non-orthogonal multiple access (NOMA) technology, the PLP model accurately modeled road and vehicle distributions, thereby increasing the spectrum efficiency and system capacity, and supporting NOMA resource allocation strategies [30]. Additionally, the PLP played a critical role in optimizing RSU deployment, helping to determine optimal strategies, reduce investment, and improve network coverage and communication reliability [31].

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Inspired by the aforementioned works, we apply ISAC technology to vehicular networks, using the PLP to model roads and two one-dimensional PPPs to model the distribution of RSUs and vehicles in this work. By employing a time-division multiplexing strategy, we theoretically quantify the density of sensing and communication RSUs. Initially, we derive the probability of successful perception (PSP), which is subsequently correlated with the density of communicating RSUs through the resource allocation strategy, yielding the communication coverage probability (CCP) under the condition of successful sensing. Finally, we apply the conditional probability formula to calculate the JPISAC. Further details of the specific contributions will follow.

1) Modeling ISAC-enabled VCN using PLP: We use a doubly stochastic spatial model, where the road system is represented by PLP, and the locations of vehicles and RSUs are modeled with two one-dimensional PPPs. The *Swerling I model* is applied for the sensing phase, and the Nakagami-*m* fading model is assumed for the communication phase.

2) PSP: We assume that the typical RSU and vehicle are located on the same road. Based on this, interference is classified into two types: from RSUs on the same road and from RSUs on other roads. Based on this classification, we then derive the PSP using the Laplace transform of the interference power distribution.

3) CCP: We introduce a time-division multiplexing strategy for sensing and communication within the ISAC framework, linking the density of communication RSUs to the PSP. We derive the CCP via the Laplace transform of interference power, incorporating the PSP. Furthermore, we evaluate the trade-off relationship between the sensing performance (PSP) and communication performance (CCP), which arises from the challenges of resource allocation strategy and interference management.

4) JPISAC: We combine the PSP with the CCP to derive their joint probability. Since there is interdependence between sensing success and communication coverage, we use the conditional probability formula to calculate the JPISAC for the ISAC-enabled VCN.

The rest of the paper is organized as follows: Sec. III describes the system model. Sec. IV provides a detailed derivation of main performance metrics of this work. Sec. V presents the simulation and numerical results. Finally, Sec. VI concludes the paper.

III. System Model

In this section, we construct the key components of the system model within the ISAC-enabled VCN scenario. First, we characterize the spatial distribution of roads using the PLP, while the vehicles and RSUs are modeled using two one-dimensional PPPs. We then explain the resource allocation strategy, detailing the time-division multiplexing for sensing and communication, and how time slots are allocated based on sensing success or failure. Next, we describe the channel propagation characteristics, including the Nakagami-*m* fading and path loss models, to evaluate signal propagation and aggregated interference. Finally, we define the signal-to-interference ratio (SIR) for both sensing and communication.

For the ISAC-enabled VCN scenario, the main focus is the operational mechanism of the downlink wireless transmission link. The RSU first sends out radar beams to acquire the vehicle's real-time location. The radar beams hit the target vehicle and bounce back, with the RSU receiving the echo to determine its position. The communication data depends on the information detected by the radar and is transmitted only when the sensing is successful; otherwise, the system enters a silent state currently.

We model the road network using a PLP,[1] denoted by Φ_L, of which the line density μ_l represents the road density per unit area. Φ_L is a motion-invariant line process. In the representation space $C \equiv \mathbb{R}^+ \times [0, 2\pi]$, each line is characterized by its orientation angle and its perpendicular distance from the origin. Based on this representation, the corresponding point process density is given by $\lambda_l = \frac{\mu_l}{\pi}$ [1]. The RSUs on the roads are modeled as a one-dimensional PPP Φ_r with density λ_r, which represents the RSU count per unit road length. We focus on the network performance for a typical communication and sensing pair, where the typical vehicle is located at the origin o, assuming that the typical RSU and the typical vehicle are located on the same road, with the distance denoted as d. Since the typical vehicle must be located on a road, we introduce a road L_0 passing through the origin, referred to as the typical road. Applying Slivnyak's theorem, the resulting line process is given by $\Phi_{L_0} = \Phi_L \cup L_0$. Subsequently, we define y_i as the relative distance along L_0 from the i-th interfering RSU, and x_i as the projected distance along L_i. The interference distance is calculated using the Euclidean formula $\sqrt{{x_i}^2 + {y_i}^2}$, with its range being $(0, +\infty)$. To fully describe the system's geometric configuration, the counterclockwise angle from the x-axis to L_0 is set as θ. The overall setup is illustrated in Fig. 1, which depicts the spatial relationship between the typical vehicle, the typical RSU, and potential interfering RSUs within the ISAC-enabled VCN. We have designed a time-division multiplexing framework for sensing and communication in ISAC-enabled VCN. This framework organizes time into ISAC frames, where time slots are dynamically allocated based on the success of sensing. We use a threshold-based echo detection method, where the PSP P_s represents the probability that the SIR at the current time exceeds the predefined threshold τ, and the sensing process is considered successful when this condition is met. The total time T represents all the time slots in an ISAC frame. The ISAC frame time is divided into a fixed sensing phase T_s and a communication phase, both operating at the same power level P. Whether the communication phase is used for data transmission T_c or enters a silent phase T_d depends on the outcome of the sensing process. If sensing is successful, the total time is represented as $T = T_s + T_c$. If sensing fails, the RSU will enter a silent phase T_d during the communication phase to avoid redundant interference with other communications, in which case the total time is represented as $T = T_s + T_d$. In this case, the communication phase is not used for data transmission, and the total time

[1]The PLP is a mathematically tractable model widely used for representing irregular road topologies in VCN and communication systems. While it does not explicitly characterize grid-like structures, it provides an effective statistical approximation in large-scale networks with directional heterogeneity.

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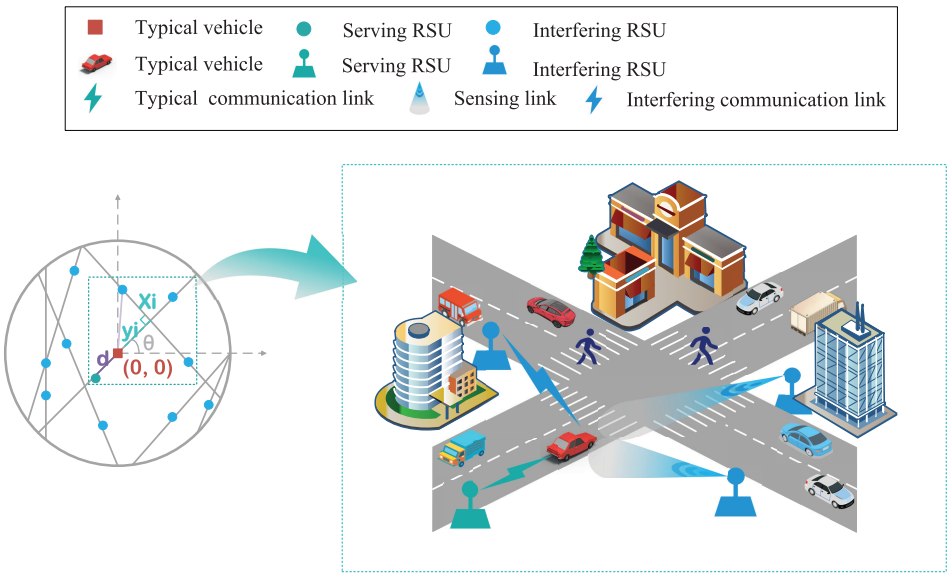

Fig. 1. Illustration of the system model.

remains T. The ratio $\beta = \frac{T_s}{T}$ serves as a time resource allocation parameter, balancing the time between sensing and communication to ensure reliable sensing.

In the propagation model for ISAC-enabled VCN, vehicles are equipped with omnidirectional antennas with a gain of G_v, allowing them to receive signals uniformly from all directions. The RSU is equipped with two sets of directional antenna arrays, one for sensing and the other for communication. The sensing antenna is responsible for both transmission and reception of signals. The main lobe gain of the sensing transmitting antenna is M_s^T, the side lobe gain is m_s^T, and the beam width is θ_s. The main lobe of the transmitting antenna at the typical RSU is always pointed toward the typical vehicle, while the probability that the main lobe of the interfering RSU's transmitting antenna causes interference to the typical vehicle is $q_s^T = \frac{\theta_s}{2\pi}$, and the probability of interference through the side lobe is $1 - q_s^T$. Similarly, the main lobe gain of the sensing receiving antenna is M_s^R, and the side lobe gain is m_s^R. During signal receiving, the probability that the interfering RSU's signal enters the main lobe of the receiving antenna is $q_s^R = q_s^T$, and the probability of interference through the side lobe is $1 - q_s^R$. We use the *Swerling I model*[2] to describe the fluctuations of the target's RCS σ. The probability density function (PDF) for any non-negative σ is given by $P(\sigma) = \frac{1}{\bar{\sigma}} e^{-\frac{\sigma}{\bar{\sigma}}}$, with $\bar{\sigma}$ representing the average RCS. Based on this model, the echo power received by the RSU from the target during the sensing phase can be expressed using the following formula from [34]:

$$P_e = \frac{P M_s^T M_s^R \lambda_\omega^2 \sigma}{(4\pi)^3 d^{2\alpha}}, \tag{1}$$

[2]The Swerling I model is used to describe the fluctuations of radar target radar cross section (RCS). In this model, the RCS remains constant during each scan but fluctuates between different scans. These fluctuations are represented by independent random variables that follow an exponential distribution, with the phase of each pulse independently and uniformly distributed. The Swerling I model is widely applied to describe slow RCS fluctuations of targets, particularly in complex environments such as automotive radar systems [32], [33].

where P represents the transmitting power, and $\lambda_\omega = \frac{1}{f}$ is the wavelength of the sensing signal. The RCS of the target is denoted by σ. The distance between the RSU and the target is represented by d, and α denotes the path loss exponent.

For the directional antenna array used for communication, the main lobe gain of the communication transmitting antenna is M_c^T, the side lobe gain is m_c^T, and the beam width is θ_c. During the communication phase, the main lobe of the typical RSU's transmitting antenna is always pointed toward the typical vehicle, while the probability that the main lobe of the interfering RSU's transmitting antenna causes interference to the typical vehicle is $q_c^T = \frac{\theta_c}{2\pi}$, and the probability of interference through the side lobe is $1 - q_c^T$. We assume that the wireless transmission channel experiences Nakagami-*m* fading and path loss.[3] That is to say, the fading gain G is a random variable following the Gamma distribution, with its PDF given by $f_G(g) = \frac{N^N g^{N-1} e^{-Ng}}{\Gamma(N)}$, where N is the shape parameter and $\Gamma(N)$ is the Gamma function. Additionally, the path loss model is expressed as $r^{-\alpha}$, where r is the transmission distance of the wireless link.

We denote I_s and I_c as the interference caused by sensing RSUs and communicating RSUs, respectively. Under the system settings, the SIR for typical detection is defined as:

$$\gamma_s = \frac{P_e}{I_s + I_c}. \tag{2}$$

Specifically, I_s represents the interference experienced by the typical RSU from other sensing RSUs during the sensing phase of the typical link, which is given by:

$$I_s = \underbrace{\sum_{x_i \in \Phi_s \cap \{L_0/(d\cos\theta_1, d\sin\theta_1)\}} A_s P_d \sigma l_i^{-\alpha}}_{I_s^{\text{intra}}}$$

[3]Nakagami-*m* fading has a broad range of excellent applications, as it can accurately capture the channel gain characteristics across a wide range of frequencies, which is obviously suitable for modeling ISAC channels.

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$$+\underbrace{\sum_{L_j\in\Phi_s\cap\{\Phi_{L_0}/L_0\}}\sum_{x_i\in L_j} A_s P_d \sigma l_{ji}^{-\alpha}}_{I_s^{\text{inter}}}. \tag{3}$$

Note that in Eq. (3), I_s^{intra} represents the interference power from the sensing RSUs located on the typical road, while I_s^{inter} represents the interference power from all the sensing RSUs located on the other roads. Φ_s represents the sensing RSUs, and through an independent thinning operation on the original PPP Φ_r, it forms a homogeneous PPP with a density of $\lambda_s = \beta\lambda_r$. The effective aperture of the radar receiver is given by $A_s = \frac{M_r\lambda_w^2}{4\pi}$, where M_r represents the receiving antenna gain of the typical RSU. From a statistical perspective, the value of M_r follows a probability distribution, where it equals M_s^R with probability q_s^R, and equals m_s^R with probability $1-q_s^R$. The power density at a unit distance from the interferer is expressed as $P_d = \frac{PM_t}{4\pi}$, where P is the transmitting power, and M_t represents the transmitting antenna gain of the interfering RSU. Similarly, the value of M_t is M_s^T with probability q_s^T, and m_s^T with probability $1-q_s^T$.

Subsequently, I_c represents the interference experienced by the typical vehicle from other communicating RSUs during the sensing phase of the typical link, which is given by:

$$I_c = \underbrace{\sum_{x_i\in\Phi_c\cap\{L_0/(d\cos\theta_1, d\sin\theta_1)\}} PC_tC_rG_{l_i}l_i^{-\alpha}}_{I_c^{\text{intra}}} + \underbrace{\sum_{L_j\in\Phi_c\cap\left\{\Phi_{L_0}/L_0\right\}}\sum_{x_i\in L_j} PC_tC_rG_{l_{ji}}l_{ji}^{-\alpha}}_{I_c^{\text{inter}}}. \tag{4}$$

In Eq. (4), I_c^{intra} represents the interference power from the communicating RSUs located on the typical road, while I_c^{inter} represents the interference power from all the communicating RSUs located on the other roads. The transmitting antenna gain of the interfering RSU, denoted as C_t, follows a probabilistic distribution, being M_c^T with probability q_c^T and m_c^T with probability $1-q_c^T$. Similarly, the receiving antenna gain of the typical RSU, denoted as C_r, follows a probabilistic distribution, being M_s^R with probability q_s^R and m_s^R with probability $1-q_s^R$. Φ_c represents the communication RSUs, which also forms a homogeneous PPP. This process is derived through thinning of Φ_r with a density of $\lambda_c = (1-\beta)\lambda_r P_s$. Additionally, G_{l_i} denotes the interference channel gain, $l_i^{-\alpha}$ accounts for the path loss, and l_i refers to the Euclidean distance between the i-th interfering RSU and the typical vehicle.

The SIR of typical communication is given by:

$$\gamma_c = \frac{PM_c^T G_v G_d d^{-\alpha}}{I_s + I_c}, \tag{5}$$

where G_d is the channel fading gain between the typical receiver and the typical RSU. The primary interferers include sensing RSUs, whose interference power is denoted as I_s, and communication RSUs, whose interference power is denoted as I_c. The calculation of these interferences follows the previously defined equations, Eq. (3) and Eq. (4), albeit with differences in antenna gains. From a statistical perspective, for the interference from sensing RSUs, the transmitting antenna gain C_t follows a probability distribution, being M_s^T with probability q_s^T and m_s^T with probability $1-q_s^T$, while the receiving antenna gain C_r is the omnidirectional antenna gain G_v of the typical vehicle. Similarly, for the interference from communication RSUs, the transmitting antenna gain C_t also follows a probability distribution, being M_c^T with probability q_c^T and m_c^T with probability $1-q_c^T$, with the receiving antenna gain C_r remaining the omnidirectional antenna gain G_v of the typical vehicle.

The notation used in next paper is summarized in Table I. Based on the system model proposed in this section, we will further extract the characteristics of various variables to evaluate the system's performance, which will be presented in the following section.

IV. Performance Analysis of ISAC-Enabled VCN

Based on the system model presented in the previous section, the PSP in the ISAC-enabled VCN is computed in this section. Subsequently, the CCP is evaluated, followed by an analysis of the trade-off relationship between the sensing and communication functionalities. Finally, the JPISAC is derived, providing a more comprehensive assessment of the network's sensing and communication performance within the ISAC framework.

A. Laplace Transform of the Interference

In this subsection, we derive the Laplace transform of the distribution of aggregate interference power at the typical vehicle for further analysis of both sensing and communication procedures. As mentioned previously, there are two types of interference, which are analyzed independently. We characterize the interference from each component separately. Specifically, *Lemma 1* to *Lemma 4* characterize the interference power during the sensing phase for analyzing the PSP, while *Lemma 5* to *Lemma 8* characterize the interference power during the communication phase for evaluating the CCP.

Lemma 1: The Laplace transform of the sensing interference power from the RSUs located on the typical road for the PSP is given by:

$$\mathcal{L}_{I_s^{\text{intra}}}^s(s) = \exp\left\{-2\lambda_s\sum_{i=1}^{4} P^s(a_i^s, b_i)F_1^s(a_i^s, b_i)\right\}, \tag{6}$$

where $F_1^s(a_i^s, b_i) = \int_0^\infty \frac{sPa_i^s b_i\lambda_\omega^2\bar{\sigma}x^{-\alpha}}{(4\pi)^2+sPa_i^s b_i\lambda_\omega^2\bar{\sigma}x^{-\alpha}}\,dx$. The corresponding values of $P^s(a_i^s, b_i)$ are shown in the TABLE II.

Proof: Please refer to Appendix A.■

Next, we will analyze the interference power from interferers on the other roads.

Lemma 2: The Laplace transform of the sensing interference power from the RSUs that are not located on the typical road for the PSP is given by the following formula:

$$\mathcal{L}_{I_s^{\text{inter}}}^s(s) = \exp\Big\{-2\pi\lambda_l \int_0^\infty 1-\exp\left[-2\lambda_s\sum_{i=1}^{4} P^s(a_i^s, b_i)F_2^s(a_i^s, b_i)\right]dy\Big\}, \tag{7}$$

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TABLE I
SUMMARY OF ACRONYMS AND NOTATIONS

Acronym	Full Term	Acronym	Full Term
ISAC	Integrated sensing and communication;	VCN	Vehicular communication networks;
ITS	Intelligent transportation systems;	RSUs	Roadside units;
V2I	Vehicle-to-infrastructure;	SIR	Signal-to-interference ratio;
PLP	Poisson line process;	PPP	Poisson point process;
PCP	Poisson cluster process;	PSP	Probability of successful perception;
CCP	Communication coverage probability;	JPISAC	Joint probability of ISAC coverage;
LOS	Line-of-sight;	NLOS	Non-line-of-sight;
NOMA	Non-orthogonal multiple access;	V2V	Vehicle-to-vehicle;
RCS	Radar cross section;	PDF	Probability density function;
MGF	Moment generating function;	PGFL	Probability generating functional;
PMF	Probability mass function.		

Notation	Description
$\Phi_L; \Phi_{L_0}$	PLP; A line process Φ_L that includes a line passing through the origin $o \equiv (0,0)$.
$\lambda_l; \mu_l$	Line density of PLP; Equivalent PPP density of Φ_l in space C.
$\lambda_r; \lambda_s; \lambda_c$	Density of all RSUs; Density of sensing RSUs; Density of communication RSUs.
Φ_r	PPP of RSUs.
d	Distance between the typical RSU and vehicle.
$y_i; x_i$	Vertical distance from the i-th interfering RSU to the road; Projection distance from the i-th interfering RSU to the road.
L_0	Line through (0, 0) (the typical road).
θ	The counterclockwise angle from the x-axis to L_0.
$T; T_s; T_c; T_d$	Total time in an ISAC frame; Sensing phase duration; Communication phase duration; Silent phase if sensing fails.
β	The time allocation ratio between sensing and total time.
G_v	The omnidirectional antenna gain of the vehicles.
$M_s^T; M_s^R$	Main lobe gain of the sensing transmitting antenna; Main lobe gain of the sensing receiving antenna.
$m_s^T; m_s^R$	Side lobe gain of the sensing transmitting antenna; Side lobe gain of the sensing receiving antenna.
q_s^T	Probability of interference in the main lobe of the sensing transmitting beam.
q_s^R	Probability of interference in the main lobe of the sensing receiving beam.
σ	Radar cross section of the target.
P	The transmitting power.
$M_t; M_r$	Gain of the sensing transmitting antenna; Gain of the sensing receiving antenna.
λ_ω	Wavelength of the sensing signal.
$M_c^T; m_c^T$	Main lobe gain of the communication transmitting antenna; Side lobe gain of the communication transmitting antenna.
q_c^T	Probability of interference in the main lobe of the communication transmitting beam.
α	The path loss exponent.
$\theta_s; \theta_c$	Beam width of the sensing transmitting antenna; Beam width of the communication transmitting antenna.
$G; N$	Fading gain of the channel (Gamma distribution); Shape parameter of Nakagami-*m* fading.
r	The transmission distance of the wireless link.
$l_i; G_{l_i}$	Euclidean distance between the i-th interfering RSU and the typical vehicle; Interference channel gain.
$C_t; C_r$	Gain of the communication transmitting antenna; Gain of the communication receiving antenna.
$\tau; \varepsilon$	Threshold of typical sensing SIR; Threshold of typical communication SIR.
$P_e; P_s; P_{c\|s}; P_j$	The echo power; PSP; CCP; JPISAC.

TABLE II
THE CORRESPONDING VALUES OF $P^s(a_i^s, b_i)$

a_i^s	b_i	$P^s(a_i^s, b_i)$
$a_1^s = M_s^T$	$b_1 = M_s^R$	$q_s^T q_s^R$
$a_2^s = m_s^T$	$b_2 = M_s^R$	$(1-q_s^T)q_s^R$
$a_3^s = M_s^T$	$b_3 = m_s^R$	$q_s^T(1-q_s^R)$
$a_4^s = m_s^T$	$b_4 = m_s^R$	$(1-q_s^T)(1-q_s^R)$

TABLE III
THE CORRESPONDING VALUES OF $P^c(a_i^c, b_i)$

a_i^c	b_i	$P^c(a_i^c, b_i)$
$a_1^c = M_c^T$	$b_1 = M_s^R$	$q_c^T q_s^R$
$a_2^c = m_c^T$	$b_3 = M_s^R$	$(1-q_c^T)q_s^R$
$a_3^c = M_c^T$	$b_3 = m_s^R$	$q_c^T(1-q_s^R)$
$a_4^c = m_c^T$	$b_4 = m_s^R$	$(1-q_c^T)(1-q_s^R)$

where $F_2^s(a_i^s, b_i) = \int_0^\infty \frac{sPa_i^s b_i \lambda_\omega^2 \bar{\sigma}(x^2+y^2)^{-\alpha/2}}{(4\pi)^2 + sPa_i^s b_i \lambda_\omega^2 \bar{\sigma}(x^2+y^2)^{-\alpha/2}}\, dx$.

Proof: Please refer to *Appendix B*.■

After analyzing the impact of sensing interference power from the RSUs, we now dive in the analysis of the communication interference power.

Lemma 3: For the communication interference power originating from RSUs located on the typical road, the corresponding Laplace transform for the PSP can be derived using the following formula:

$$\mathcal{L}_{I_c^{\text{intra}}}^s(s) = \exp\left\{-2\lambda_c \sum_{i=1}^{4} P^c(a_i^c, b_i) F_1^c(a_i^c, b_i)\right\}, \tag{8}$$

where $F_1^c(a_i^c, b_i) = \int_0^\infty 1 - \left(1 + \frac{sPa_i^c b_i}{mx^\alpha}\right)^{-m} dx$. The corresponding values of $P^c(a_i^c, b_i)$ are summarized in Table III.

Proof: Please refer to *Appendix C*.■

Next, we analyze the interference from RSUs located outside the typical road.

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Lemma 4: The Laplace transform of the communication interference power, which originates from RSUs situated outside the typical road is given by:

$$\mathcal{L}^{s}_{I^{\text{inter}}_{c}}(s) = \exp\Big\{ -2\pi\lambda_l \int_0^\infty 1 - \exp\left[-2\lambda_c \sum_{i=1}^{4} P^c(a_i^c, b_i) F_2^c(a_i^c, b_i)\right] dy \Big\}, \quad (9)$$

where $F_2^c(a_i^c, b_i) = \int_0^\infty 1 - \left(1 + \frac{sPa_i^c b_i}{m(x^2+y^2)^{\alpha/2}}\right)^{-m} dx$.

Proof: Please refer to *Appendix D*.■

The interference power affecting the typical link during the sensing phase has already been analyzed. We proceed to evaluate the interference power received during the communication phase. Specifically, *Lemmas 5* and *6* address the analysis of sensing interference, while *Lemmas 7* and *8* focus on the communication interference power. The proofs of *Lemmas 5* to *8* are similar to those of *Lemmas 1* to *4*, respectively, and thus the detailed proof processes are omitted.

Lemma 5: The Laplace transform of the sensing interference power generated by RSUs situated on the typical road can be expressed as:

$$\mathcal{L}^{c}_{I^{\text{intra}}_{s}}(s) = \exp\left[-2\lambda_s q_s^T F_1^s(M_s^T, G_v)\right] \times \exp\left[-2\lambda_s (1-q_s^T) F_1^s(m_s^T, G_v)\right]. \quad (10)$$

Lemma 6: The Laplace transform corresponding to the sensing interference power from RSUs outside the typical road is given by:

$$\mathcal{L}^{c}_{I^{\text{inter}}_{s}}(s) = \exp\Big\{ -2\pi\lambda_l \int_0^\infty 1 - \exp\left[-2\lambda_s q_s^T F_2^s(M_s^T, G_v)\right] \exp\left[-2\lambda_s (1-q_s^T) F_2^s(m_s^T, G_v)\right] dy \Big\}. \quad (11)$$

Lemma 7: The Laplace transform of the distribution of communication interference power from the RSUs located on the typical road for CCP is given by:

$$\mathcal{L}^{c}_{I^{\text{intra}}_{c}}(s) = \exp\left[-2\lambda_c q_c^T F_1^c(M_c^T, G_v)\right] \times \exp\left[-2\lambda_c (1-q_c^T) F_1^c(m_c^T, G_v)\right]. \quad (12)$$

Lemma 8: The communication interference power from RSUs situated outside the typical road can be represented through its Laplace transform as:

$$\mathcal{L}^{c}_{I^{\text{inter}}_{c}}(s) = \exp\Big\{ -2\pi\lambda_l \int_0^\infty 1 - \exp\left[-2\lambda_c q_c^T F_2^c(M_c^T, G_v)\right] \times \exp\left[-2\lambda_c (1-q_c^T) F_2^c(m_c^T, G_v)\right] dy \Big\}. \quad (13)$$

We have derived the Laplace transforms of the interference power for both the sensing and communication phases. Next, we will proceed with an in-depth analysis of the PSP.

B. The Probability of Successful Perception

The PSP has been formally defined as the probability that the SIR at the typical vehicle of the typical link exceeds a predetermined threshold β. Using the Laplace transform of the interference power distribution, the PSP is computed in the following theorem.

Theorem 1: The PSP P_s of the typical link is given by the following expression, where ξ is defined as $\frac{(4\pi)^3\tau}{PM_s^T M_s^R \lambda_w^2}$.

$$P_s = \mathcal{L}^{s}_{I^{\text{intra}}_{s}}\left(\frac{\xi d^{2\alpha}}{\bar{\sigma}}\right) \mathcal{L}^{s}_{I^{\text{inter}}_{s}}\left(\frac{\xi d^{2\alpha}}{\bar{\sigma}}\right) \times \mathcal{L}^{s}_{I^{\text{intra}}_{c}}\left(\frac{\xi d^{2\alpha}}{\bar{\sigma}}\right) \mathcal{L}^{s}_{I^{\text{inter}}_{c}}\left(\frac{\xi d^{2\alpha}}{\bar{\sigma}}\right), \quad (14)$$

by substituting Eqs. (6), (7), (8), and (9) into Eq. (14), the PSP is computed.

Proof: Please refer to *Appendix E*.■

C. Communication Coverage Probability

CCP refers to the likelihood that the SIR at the typical vehicle exceeds the communication threshold ε. In the ISAC-enabled VCN, communication only begins after a successful sensing, making the CCP dependent on sensing success. By leveraging the conditional Laplace transform of the interference power distribution, we compute the CCP for a typical vehicle given a successful sensing. Detailed calculation methods will be provided in the proof of the subsequent theorem.

Theorem 2: The CCP for a typical vehicle given successful sensing is

$$P_{c|s} = \sum_{n=1}^{N} (-1)^{n+1} \binom{N}{n} \Big[\mathcal{L}^{c}_{I^{\text{intra}}_{s}}\left(\frac{n\eta\varepsilon}{PM_c^T G_v d^{-\alpha}}\right) \mathcal{L}^{c}_{I^{\text{inter}}_{s}}\left(\frac{n\eta\varepsilon}{PM_c^T G_v d^{-\alpha}}\right) \mathcal{L}^{c}_{I^{\text{intra}}_{c}}\left(\frac{n\eta\varepsilon}{PM_c^T G_v d^{-\alpha}}\right) \mathcal{L}^{c}_{I^{\text{inter}}_{c}}\left(\frac{n\eta\varepsilon}{PM_c^T G_v d^{-\alpha}}\right)\Big]. \quad (15)$$

By inserting Eqs. (10), (11), (12), and (13) into Eq. (15), the PSP is calculated.

Proof: Please refer to *Appendix F*.■

Remark 1: Within ISAC-enabled VCN, the sensing performance (PSP) and communication performance (CCP) exhibit a trade-off relationship. From the formula mentioned earlier, it is evident that the density of communication RSUs is directly proportional to the PSP. When the PSP increases, the higher RSU density raises the interference power along the typical link during the communication phase, which in turn reduces the CCP. Conversely, a decrease in PSP lowers the RSU density, reducing interference and improving communication coverage. For further validation, detailed simulations and numerical analyses will be presented in the next section.

Building on the analysis in Sec. IV-B and C, we can further combine the metric from the sensing and communication phases to derive the JPISAC. This metric reflects the integrated performance of the network and reveals the trade-off relationship between sensing and communication.

D. The Joint Probability of ISAC Coverage

The JPISAC refers to the probability that both successful sensing and communication coverage occur, and *Theorem 3* presents the detailed mathematical formulation of this probability.

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TABLE IV
EXAMPLE PARAMETER SETTINGS

Parameter	Value	Parameter	Value
P	10 dBm	$\bar{\sigma}$	30 dBsm
M_s^T	30 dBi	M_c^T	30 dBi
m_s^T	0.63	m_c^T	0.63
M_s^R	30 dBi	M_c^R	30 dBi
m_s^R	0.63	m_c^R	0.63
f	76.5 GHz		

Theorem 3: We present the joint probability formula, which describes the probability that both successful sensing and communication coverage occur.

$$P_j = \mathcal{L}^s_{I_s^{\text{intra}}}\left(\frac{\xi d^{2\alpha}}{\bar{\sigma}}\right)\mathcal{L}^s_{I_s^{\text{inter}}}\left(\frac{\xi d^{2\alpha}}{\bar{\sigma}}\right)$$
$$\mathcal{L}^s_{I_c^{\text{intra}}}\left(\frac{\xi d^{2\alpha}}{\bar{\sigma}}\right)\mathcal{L}^s_{I_c^{\text{inter}}}\left(\frac{\xi d^{2\alpha}}{\bar{\sigma}}\right)\sum_{n=1}^{N}(-1)^{n+1}\binom{N}{n}$$
$$\left[\mathcal{L}^c_{I_s^{\text{intra}}}\left(\frac{n\eta\varepsilon}{PM_c^T G_v d^{-\alpha}}\right)\mathcal{L}^c_{I_s^{\text{inter}}}\left(\frac{n\eta\varepsilon}{PM_c^T G_v d^{-\alpha}}\right)\right.$$
$$\left.\mathcal{L}^c_{I_c^{\text{intra}}}\left(\frac{n\eta\varepsilon}{PM_c^T G_v d^{-\alpha}}\right)\mathcal{L}^c_{I_c^{\text{inter}}}\left(\frac{n\eta\varepsilon}{PM_c^T G_v d^{-\alpha}}\right)\right], \quad (16)$$

where the expressions from Eq. (6) to Eq. (13) are substituted into Eq. (16) to obtain the JPISAC.

Proof: Please refer to *Appendix G*.■

This section analyzes the PSP and CCP in ISAC-enabled VCN and derives the JPISAC. The study reveals that the trade-off relationship between PSP and CCP provides insights for efficient resource allocation under resource constraints. These findings offer theoretical support for resource management and system performance improvements in complex network environments, laying the foundation for future system design and optimization.

V. SIMULATIONS AND NUMERICAL RESULTS

Based on the theoretical analysis provided in the previous section, we conduct simulations and numerical experiments to validate the results. Also, through the numerical results, we evaluate the model's performance under different parameter settings, with the default system settings provided in TABLE IV. Except for the side-lobe gain, all default parameters are selected based on the recommendations in the technical report by the International Telecommunication Union [34] for automotive radar systems operating in the 76-81 GHz band. The side-lobe gain is chosen with reference to both engineering practices and prior literature: on the one hand, [35] provides modeling insights for side-lobe interference under transient antenna misalignment; on the other hand, [36] states that the radiation intensity in the side-lobe direction is typically at least 30 dB lower than that in the main-lobe direction, thereby supporting the engineering rationality of our adopted side-lobe gain value of 0.63. In all plots, we observe that the model's predictions closely align with the theoretical values, which verifies the accuracy and robustness of the proposed model.

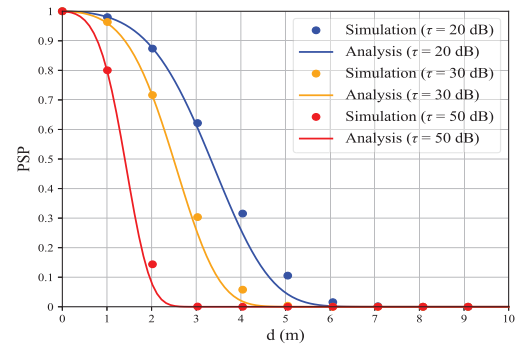

Fig. 2. PSP versus service distance d.

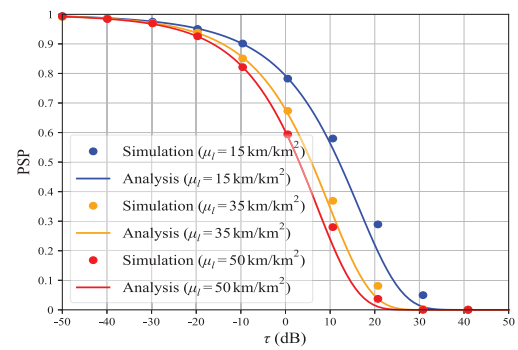

Fig. 3. PSP versus threshold τ for different line densities μ_l.

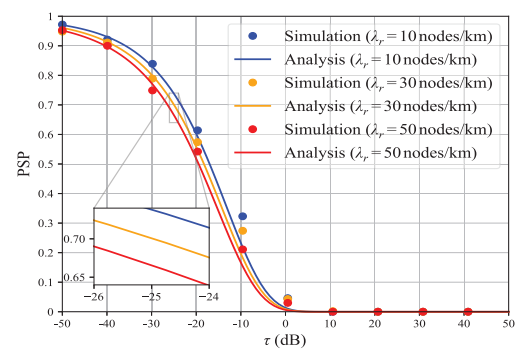

Fig. 4. PSP versus threshold τ for different RSU densities λ_r.

First, we plot Figs. 2, 3, and 4 to evaluate the PSP under different distance d, threshold τ, RSU density λ_r and line density μ_l. These figures reveal several key trends. As shown in Fig. 2, the PSP decreases as the d increases. This is primarily due to the reduced echo power at longer distances. Furthermore, a higher τ exacerbates this decline, as demonstrated in *Lemma 9*, where the PSP is a decreasing function of τ, consistent with the properties of the cumulative distribution function (CDF). Figs. 3 and 4 further validate this trend. In Fig. 3, we observe that increasing μ_l leads to a decrease in the PSP. This is mainly because a higher μ_l increases the number of interfering RSUs, thus raising the interference power. Fig. 4 also illustrates the inverse relationship between PSP and λ_r. As the value of λ_r increases, the average distance between interfering RSUs and the typical vehicle decreases, which leads to the higher interference power and consequently reduces the PSP.

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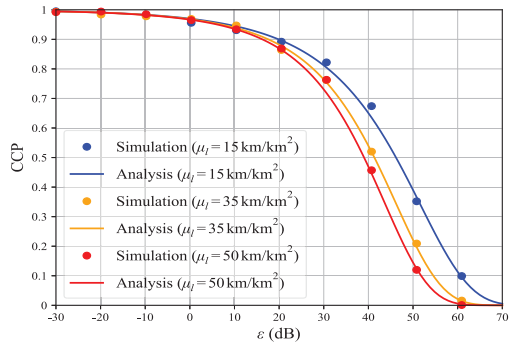

Fig. 5. CCP versus threshold ε for different line densities μ_l.

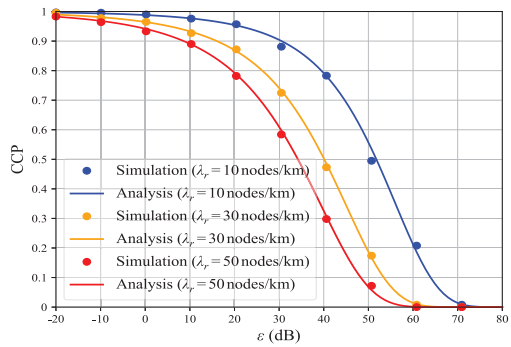

Fig. 6. CCP versus threshold ε for different RSU densities λ_r.

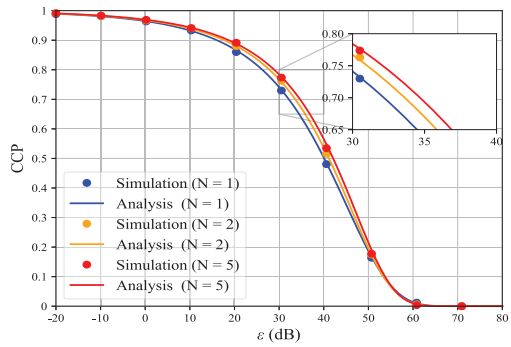

Fig. 7. CCP versus threshold ε for different Nakagami-m fading shape parameters N.

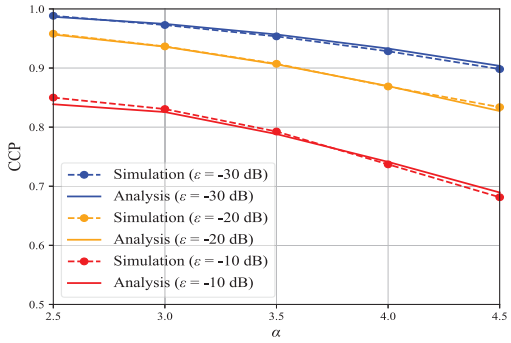

Fig. 8. CCP versus path-loss exponent α for various thresholds ε.

We evaluate the impact of various system parameters on CCP, including the threshold ε, line density μ_l, RSU density λ_r, Nakagami-m fading shape parameter N and the path-loss exponent α. Figs. 5, 6, 7, and 8 all reveal a significant inverse relationship between CCP and the ε. This observation closely aligns with the conclusion of *Lemma10* and is fully consistent with the general properties of the CDF. Fig. 5 further indicates that an increase in μ_l leads to a decrease in CCP. This is because higher μ_l introduces more interferers, increasing the interference power and thus reducing CCP. For a fixed $\mu_l = 35\,\text{km/km}^2$, Fig. 6 shows that CCP decreases as λ_r increases. This occurs because a higher λ_r shortens the average interfering distance, thus increasing interference power. In Fig. 7, we analyze the impact of N on $P_{c|s}$. As N increases, the CCP also increases. A larger shape parameter N indicates that the signal fading is closer to a Gaussian distribution, meaning that the signal strength becomes less variable. This stabilizes the signal, reduces interference, and consequently improves the CCP. Finally, Fig. 8 shows that the CCP of a typical vehicle is inversely proportional to the α.

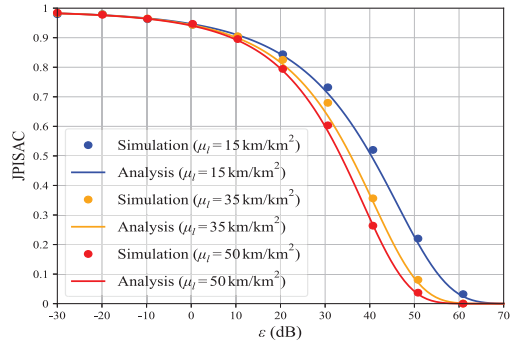

Fig. 9. JPISAC versus threshold for different line densities μ_l.

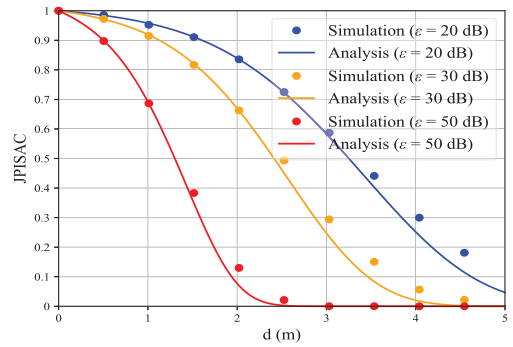

Fig. 10. JPISAC versus service distance d for various thresholds.

Building on the previous analysis, we now evaluate how different system parameters, such as the threshold, line density μ_l, and service distance d, affect the JPISAC. Figs. 9 and 10 show that the JPISAC is inversely related to threshold, which is consistent with the conclusion of Eq. (16) that the JPISAC is a decreasing function of threshold. This aligns with the general properties of the CDF. Fig. 9 shows that as μ_l increases, the JPISAC decreases. A higher μ_l shortens the interfering distance and increases the interference power. As a result, both PSP and CCP are reduced, leading to a lower JPISAC. Fig. 10 illustrates the relationship between the JPISAC and d. It can be observed that the JPISAC decreases significantly as

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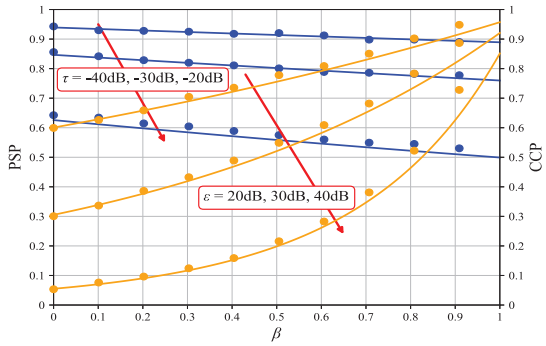

Fig. 11. PSP and CCP versus β. Blue lines and dots denote analytical and simulation results of PSP, respectively; orange lines and dots denote those of CCP.

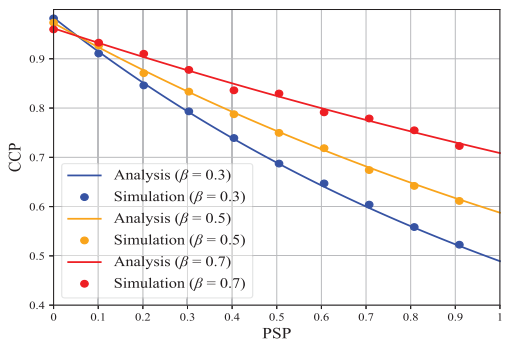

Fig. 12. CCP versus PSP for different values of the ratio β.

d increases. This is due to the reduction in PSP and CCP with increasing d, which lowers the JPISAC.

Finally, we explore the comparative impact of the system parameter ratio β and the threshold on sensing performance (PSP) and communication performance (CCP). Figs. 11 and 12 visually demonstrate the trade-off relationship between PSP and CCP. As shown in Fig. 11, both PSP and CCP decrease as the threshold increases, consistent with the previous analysis. Additionally, as β increases, PSP decreases while CCP shows an increasing trend. This can be explained by the formula $\lambda_s = \beta\lambda_r$: as β increases, the density of sensing RSUs increases, leading to more interferers and thus reducing PSP. On the other hand, the formula $\lambda_c = (1-\beta)\lambda_r P_s$ indicates that as β increases, the decrease in PSP reduces the density of communication RSUs. This increase in the interfering distance consequently improves the CCP. Fig. 12 further analyzes the impact of β on CCP. It can be observed that CCP is inversely related to PSP, consistent with the analysis in Fig. 11. As β increases, the rate of decrease in CCP slows down. According to the formula $\lambda_c = (1-\beta)\lambda_r P_s$, the derivative of λ_c with respect to P_s is $\lambda_c' = (1-\beta)\lambda_r$. As β increases, this derivative decreases, indicating that the rate of decrease in λ_c slows down, which in turn results in a slower decline in CCP.

Through simulations and numerical experiments, the system's performance has been thoroughly evaluated. The results not only validate the theoretical analysis, but also provide valuable insights for the design of ISAC-enabled VCN, particularly in terms of network deployment, resource allocation, and balancing the trade-off between the sensing performance (PSP) and communication performance (CCP).

VI. Conclusion

In this work, we use a PLP-based network model to analyze the JPISAC for a typical link in the ISAC-enabled VCN. We begin by analyzing the interference power received during the sensing phase, utilizing time-division multiplexing between the sensing and communication phases. From this analysis, we derive the expression for the PSP. Then, under the condition of successful sensing, we further analyze the CCP during the communication phase. Based on the above analysis, we also explore the trade-off relationship between sensing and communication, highlighting the relationship between PSP and CCP. Finally, the JPISAC is derived using the conditional probability formula. Simulations validated the analysis, and the conclusions provided valuable insights for the ISAC-enabled VCN.

Appendix

A. Proof of Lemma 1

We now present the proof by expressing the Laplace transform of the interference power on the typical road as follows.

$$\begin{aligned}
&\mathcal{L}^s_{I_s^{\text{intra}}}(s) = E_{I_s^{\text{intra}}}\left[\exp\left(-sI_s^{\text{intra}}\right)\right] \\
&= E_{M_t,M_r,\sigma,x_i}\left[\prod_{x_i}\exp\left(-\frac{sPM_tM_r\lambda_\omega^2\sigma l_i^{-\alpha}}{(4\pi)^2}\right)\right] \\
&\overset{(a)}{=} E_{x_i}\left[\prod_{x_i}E_{M_t,M_r}\left(\frac{1}{1+\frac{sPM_tM_r\lambda_\omega^2\bar{\sigma}l_i^{-\alpha}}{(4\pi)^2}}\right)\right] \\
&\overset{(b)}{=} E_{M_t,M_r}\left[\exp\left(-2\lambda_s F_1^s(M_t,M_r)\right)\right] \\
&\overset{(c)}{=} \exp\Big\{-2\lambda_s\Big[q_s^T q_s^R F_1^s(M_s^T,M_s^R) \\
&\quad + q_s^T(1-q_s^R)F_1^s(M_s^T,m_s^R) + q_s^R(1-q_s^T)F_1^s(m_s^T,M_s^R) \\
&\quad + (1-q_s^T)(1-q_s^R)F_1^s(m_s^T,m_s^R)\Big]\Big\}.
\end{aligned}$$

Here, $x_i \in \{d\cos\theta, d\sin\theta\}$ represents the spatial locations of interferers. Step (a) follows the moment generating function (MGF) of the exponential variable σ, step (b) follows from the probability generating functional (PGFL) of the 1D PPP on the typical road, and step (c) follows the probability mass function (PMF) of interfering antenna gain.

B. Proof of Lemma 2

Along the lines of the proof of *Lemma 1*, we have

$$\begin{aligned}
&\mathcal{L}^s_{I_s^{\text{inter}}}(s) = \mathbb{E}_{I_s^{\text{inter}}}\left[\exp\left(-sI_s^{\text{inter}}\right)\right] \\
&= \prod_{L_j}\mathbb{E}_{M_t,M_r,L_j,x_i,\sigma}\left[\prod_{x_i\in L_j}\exp\left(-s\frac{PM_tM_r\lambda_\omega^2\sigma l_{ji}^{-\alpha}}{(4\pi)^2}\right)\right] \\
&\overset{(a)}{=} \prod_{L_j}\mathbb{E}_{L_j,x_i}\left[\prod_{x_i\in L_j}\mathbb{E}_{M_t,M_r}\left(\frac{1}{1+s\frac{PM_tM_r\lambda_\omega^2\bar{\sigma}l_{ji}^{-\alpha}}{(4\pi)^2}}\right)\right] \\
&\overset{(b)}{=} \prod_{L_j}\mathbb{E}_{L_j,M_t,M_r}\Big[\exp\Big(-2\lambda_s
\end{aligned}$$

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$$\int_0^\infty \frac{sPM_tM_r\lambda_\omega^2\bar{\sigma}(x^2+y_i^2)^{-\alpha/2}}{(4\pi)^2+sPM_tM_r\lambda_\omega^2\bar{\sigma}(x^2+y_i^2)^{-\alpha/2}}\,dx\ \Big)\Big]$$
$$\overset{(c)}{=}\prod_{L_j}\mathbb{E}_{L_j}\Big\{\exp\Big[-2\lambda_s\Big(q_s^Tq_s^R\delta(M_s^T,M_s^R)$$
$$+q_s^T(1-q_s^R)\delta(M_s^T,m_s^R)+q_s^R(1-q_s^T)\delta(m_s^T,M_s^R)$$
$$+(1-q_s^T)(1-q_s^R)\delta(m_s^T,m_s^R)\Big)\Big]\Big\}$$
$$\overset{(d)}{=}\exp\Big\{-2\pi\lambda_l\int_0^\infty 1-\exp\Big[-2\lambda_s\Big(q_s^Tq_s^RF_2^s(M_s^T,M_s^R)$$
$$+q_s^T(1-q_s^R)F_2^s(M_s^T,m_s^R)+q_s^R(1-q_s^T)F_2^s(m_s^T,M_s^R)$$
$$+(1-q_s^T)(1-q_s^R)F_2^s(m_s^T,m_s^R)\Big)\Big]dy\Big\},$$

where $L_j\in\Phi_L$ denotes all roads except the typical one. Since the interferers are located off the typical road, we need to additionally account for the distribution of the roads where the interferers are positioned. In the proof, steps (a), (b), and (c) follow exactly from the corresponding steps in *Lemma 1*. The term $\delta(a,b)$, defined as $\int_0^\infty \frac{sPab\lambda_\omega^2\bar{\sigma}(x^2+y_i^2)^{-\alpha/2}}{(4\pi)^2+sPab\lambda_\omega^2\bar{\sigma}(x^2+y_i^2)^{-\alpha/2}}\,dx$ is introduced. Furthermore, step (d) follows from the PGFL of the 2D PPP.

C. Proof of Lemma 3

The distribution of RSUs on the typical road is independent of the PLP. Therefore, the Laplace transform of the interference distribution from the RSUs on the typical road can be calculated as:

$$\mathcal{L}_{I_c^{\text{intra}}}^s(s)=\mathbb{E}_{I_c^{\text{intra}}}\left[\exp\left(-sI_c^{\text{intra}}\right)\right]$$
$$=\mathbb{E}_{x_i,C_t,C_r,G_{li}}\left[\prod_{x_i}\exp\left(-sPC_tC_rG_{li}l_i^{-\alpha}\right)\right]$$
$$\overset{(a)}{=}\mathbb{E}_{x_i}\left[\prod_{x_i}\mathbb{E}_{C_t,C_r}\left(1+\frac{sPC_tC_r}{mx^\alpha}\right)^{-m}\right]$$
$$\overset{(b)}{=}\mathbb{E}_{C_t,C_r}\left\{\exp\left[-2\lambda_c\int_0^\infty 1-\left(1+\frac{sPC_tC_r}{mx^\alpha}\right)^{-m}dx\right]\right\}$$
$$\overset{(c)}{=}\exp\Big\{-2\lambda_c\Big[q_c^Tq_s^RF_1^c(M_c^T,M_s^R)+q_s^R(1-q_c^T)$$
$$F_1^c(m_c^T,M_s^R)+q_c^T(1-q_s^R)F_1^c(M_c^T,m_s^R)$$
$$+(1-q_c^T)(1-q_s^R)F_1^c(m_c^T,m_s^R)\Big]\Big\},$$

where step (a) follows the MGF of the Gamma variable, step (b) from the PGFL of the 1D PPP on each line, and step (c) from the PMF of the interfering antenna gain.

D. Proof of Lemma 4

The positions of RSUs that do not lie on the typical road are inherently linked to the PLP, with the derivation detailed as follows:

$$\mathcal{L}_{I_c^{\text{inter}}}^s(s)=\mathbb{E}_{I_c^{\text{inter}}}\left[\exp\left(-sI_c^{\text{inter}}\right)\right]$$
$$=\prod_{L_j}\mathbb{E}_{x_i,L_j,C_t,C_r,G_{l_{ji}}}\left[\prod_{x_i\in L_j}\exp\left(-sPC_tC_rG_{l_{ji}}l_{ji}^{-\alpha}\right)\right]$$
$$\overset{(a)}{=}\prod_{L_j}\mathbb{E}_{x_i,L_j}\left[\prod_{x_i\in L_j}\mathbb{E}_{C_t,C_r}\left(1+\frac{sPC_tC_r}{ml_{ji}^\alpha}\right)^{-m}\right]$$
$$\overset{(b)}{=}\prod_{L_j}\mathbb{E}_{C_t,C_r,L_j}$$
$$\left[\exp\left(-2\lambda_c\int_0^\infty 1-\left(1+\frac{sPC_tC_r}{m(x^2+y_i^2)^{a/2}}\right)^{-m}dx\right)\right]$$
$$\overset{(c)}{=}\prod_{L_j}\mathbb{E}_{L_j}\Big\{\exp\Big[-2\lambda_c\Big(q_c^Tq_s^R\eta(M_c^T,M_s^R)$$
$$+q_s^R(1-q_c^T)\eta(m_c^T,M_s^R)+q_c^T(1-q_s^R)\eta(M_c^T,m_s^R)$$
$$+(1-q_c^T)(1-q_s^R)\eta(m_c^T,m_s^T)\Big)\Big]\Big\}$$
$$\overset{(d)}{=}\exp\Big\{-2\pi\lambda_l\int_0^\infty 1-\exp\Big[-2\lambda_c\Big(q_c^Tq_s^RF_2^c(M_c^T,M_s^R)$$
$$+q_s^R(1-q_c^T)F_2^c(m_c^T,M_s^R)+q_c^T(1-q_s^R)F_2^c(M_c^T,m_s^R)$$
$$+(1-q_c^T)(1-q_s^R)F_2^c(m_c^T,m_s^T)\Big)\Big]\,dy\ \Big\}.$$

Steps (a), (b), and (c) are consistent with the derivation in *Lemma 3*, with the key difference being in the definition of $\eta(a,b)$ and step (d). Here, $\eta(a,b)$ is given by$\int_0^\infty 1-\left(1+\frac{sPab}{m(x^2+y_i^2)^{\alpha/2}}\right)^{-m}dx$ and step (d) is derived from the PGFL for the 2D PPP representing the line process Φ_l in space C.

E. Proof of Theorem 1

To further the analysis, we proceed with the following detailed derivation:

$$P_s=\Pr\left\{\frac{PM_s^TM_s^R\lambda_\omega^2\sigma}{(4\pi)^3d^{2\alpha}(I_s+I_c)}>\tau\right\}$$
$$=\Pr\left\{\sigma>\frac{(4\pi)^3d^{2\alpha}\tau(I_s+I_c)}{PM_s^TM_s^R\lambda_\omega^2}\right\}$$
$$=1-\Pr\left\{\sigma<\xi d^{2\alpha}(I_s+I_c)\right\}$$
$$\overset{(a)}{=}\exp\left(-\frac{\xi d^{2\alpha}(I_s+I_c)}{\bar{\sigma}}\right)$$
$$=\mathbb{E}_{I_s+I_c}\left[\exp\left(-\frac{\xi d^{2\alpha}(I_s+I_c)}{\bar{\sigma}}\right)\right]$$
$$=\mathcal{L}_{I_s+I_c}\left(\frac{\xi d^{2\alpha}}{\bar{\sigma}}\right)$$
$$=\mathcal{L}_{I_s^{\text{intra}}}^s\left(\frac{\xi d^{2\alpha}}{\bar{\sigma}}\right)\mathcal{L}_{I_s^{\text{inter}}}^s\left(\frac{\xi d^{2\alpha}}{\bar{\sigma}}\right)$$
$$\times\mathcal{L}_{I_c^{\text{intra}}}^s\left(\frac{\xi d^{2\alpha}}{\bar{\sigma}}\right)\mathcal{L}_{I_c^{\text{inter}}}^s\left(\frac{\xi d^{2\alpha}}{\bar{\sigma}}\right),$$

where step (a) utilizes the CDF of the exponential distribution for the RCS σ to compute the expectation.

F. Proof of Theorem 2

Taking the expectation with respect to G_d, I_s, and I_c, we have

$$P_{c|s}=\Pr\left\{\gamma_c>\varepsilon\mid\gamma_s>\tau\right\}$$
$$=\Pr\left\{\frac{PM_c^TG_vG_dd^{-\alpha}}{I_s+I_c}>\varepsilon\mid\gamma_s>\tau\right\}$$

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$$
\begin{aligned}
&= 1 - \Pr\left\{ G_d < \frac{\varepsilon(I_s + I_c)}{PM_c^T G_v d^{-\alpha}} \mid \gamma_s > \tau \right\} \\
&\overset{(a)}{\approx} 1 - \left\{ E_{I_s+I_c}\left[1 - \exp\left(-\frac{\eta\varepsilon(I_s + I_c)}{PM_c^T G_v d^{-\alpha}} \right) \right]^N \mid \gamma_s > \tau \right\} \\
&= \sum_{n=1}^{N} (-1)^{n+1} \binom{N}{n} \\
&\quad \times \left[\mathcal{L}^c_{I_s^{\text{intra}}}\left(\frac{n\eta\varepsilon}{PM_c^T G_v d^{-\alpha}} \right) \mathcal{L}^c_{I_s^{\text{inter}}}\left(\frac{n\eta\varepsilon}{PM_c^T G_v d^{-\alpha}} \right) \right. \\
&\quad \left. \mathcal{L}^c_{I_c^{\text{intra}}}\left(\frac{n\eta\varepsilon}{PM_c^T G_v d^{-\alpha}} \right) \mathcal{L}^c_{I_c^{\text{inter}}}\left(\frac{n\eta\varepsilon}{PM_c^T G_v d^{-\alpha}} \right) \right],
\end{aligned}
$$

where in step (a), the upper bound approximation of the Gamma random variable is utilized, with $\eta = N(N!)^{-\frac{1}{N}}$ as defined in [37]. In step (b), the expressions $\mathcal{L}^c_{I_c^{\text{intra}}}\left(\frac{n\eta\varepsilon}{PM_c^T G_v d^{-\alpha}}\right)$ and $\mathcal{L}^c_{I_c^{\text{inter}}}\left(\frac{n\eta\varepsilon}{PM_c^T G_v d^{-\alpha}}\right)$ with $\lambda_c = (1-\beta)\lambda_r P_s$ reflect the CCP conditioned on successful sensing.

G. Proof of Theorem 3

Using the conditional probability formula, the following derivation is performed.

$$
\begin{aligned}
P_j &= P\left(\gamma_c > \varepsilon \cap \gamma_s > \tau\right) \\
&\overset{(a)}{=} P\left(\gamma_c > \varepsilon \mid \gamma_s > \tau\right) P\left(\gamma_s > \tau\right) \\
&= P_{c|s} P_s,
\end{aligned}
$$

where step (a) applies the conditional probability formula $P(A \cap B) = P(A \mid B)P(B)$, representing the product of the probability of achieving communication coverage, given successful sensing, and the PSP. By applying the conclusions from *Theorems 1* and *2*, the relevant expressions can be substituted to derive the final result in P_j.

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